

ŘEŠENÍ Bonus 2

①

$$a) \left(\begin{array}{ccc|c} 1 & -2 & 1 & 1 \\ 2 & 1 & -1 & 2 \\ 1 & 1 & -2 & 3 \end{array} \right) \xrightarrow{\substack{1 \cdot (-2) \quad 1 \cdot (-1) \\ \leftarrow + \\ \leftarrow +}} \sim \left(\begin{array}{ccc|c} 1 & -2 & 1 & 1 \\ 0 & 5 & -3 & 0 \\ 0 & 3 & -3 & 2 \end{array} \right) \xrightarrow{\substack{1 \cdot (-3) \\ 1 \cdot 5 \leftarrow +}}$$

$$\sim \left(\begin{array}{ccc|c} 1 & -2 & 1 & 1 \\ 0 & 5 & -3 & 0 \\ 0 & 0 & -6 & 10 \end{array} \right)$$

$$-6x_3 = 10$$

$$\underline{x_3 = -\frac{5}{3}}$$

$$5x_2 - 3x_3 = 0$$

$$5x_2 + 5 = 0$$

$$\underline{x_2 = -1}$$

$$x_1 - 2x_2 + x_3 = 0$$

$$x_1 + 2 - \frac{5}{3} = 0$$

$$\underline{x_1 = -\frac{1}{3}}$$

$$b) \begin{pmatrix} 1 & -2 & 1 \\ 2 & 1 & -1 \\ 1 & 1 & -2 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

$$\textcircled{2} \quad \begin{vmatrix} 2-\lambda & 3 \\ 1 & 4-\lambda \end{vmatrix} = (2-\lambda)(4-\lambda) - 3 = \lambda^2 - 6\lambda + 5 = (\lambda-1)(\lambda-5)$$

$$\underline{\underline{\lambda_1 = 1, \lambda_2 = 5}}$$

VLASTNÍ VEKTORY:

$$\lambda_1 = 1 : \begin{pmatrix} 1 & 3 \\ 1 & 3 \end{pmatrix} \cdot \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{matrix} u_1 + 3u_2 = 0 \\ u_1 = -3u_2 \end{matrix} \Rightarrow \underline{\underline{\begin{pmatrix} -3 \\ 1 \end{pmatrix}}}$$

$$\lambda_2 = 5 : \begin{pmatrix} -3 & 3 \\ 1 & -1 \end{pmatrix} \cdot \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{matrix} u_1 - u_2 = 0 \\ u_1 = u_2 \end{matrix} \Rightarrow \underline{\underline{\begin{pmatrix} 1 \\ 1 \end{pmatrix}}}$$